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A GWSOT Details

An agent and a goal are placed inside a grid of size 5×5 . The agent is positioned semi-randomly, ensuring that it starts at least two steps away from the goal. At each step of a trial, the agent must choose one of four possible actions—up, down, left, or right. All actions remain available even when the agent is at the grid’s boundary; if an action would move the agent outside the grid, its position is not updated. Prior to making a decision, the current world state is provided to the LLM as a user message. The full prompt given to the model is comprised of a conversation where user messages are world states and assistant messages represent past decisions. The prompt received by the LLM always starts with a user message (the initial world state) and alternates between user and assistant roles until a final user message (the current world state). This allows in-context learning, as the LLM can use information about how previous actions have altered the environment to adjust its strategy. The model is then required to output a JSON object containing a single field, "action", whose value is a string corresponding to the chosen move. Invalid responses result in the agent’s position not being updated, although these were only infrequently observed in the 1B and 3B models.

We define an action as correct if it brings the agent closer to the goal and as incorrect otherwise. A trial terminates when the agent successfully reaches the goal or when the maximum number of steps is exceeded, which is defined as $2 \times N$ (where N is the grid size). In the 5×5 grid, the agent is allowed a maximum of 10 steps. This number is chosen so that the agent can always reach the goal via a non-optimal path, even if the agent and the goal start positioned at opposite corners of the grid.

The following system prompt accompanies all representations to help the models understand the task instructions:

You are a navigation assistant tasked with guiding an Agent to a Goal in a 5x5 grid world. The Agent will start in a random position, and your objective is to provide directions that bring the Agent to the Goal.

Each time you receive an updated state of the world, choose the optimal next move to bring the Agent closer to the Goal. You may only respond with a JSON object containing a single field named "action", which should contain one of the following strings in all capital letters: "UP", "DOWN", "LEFT", or "RIGHT".

Ensure that your response is strictly in this format, with no additional text or commentary, since it will be used automatically and with no human supervision by a Python script to get the next world state.

Remember: